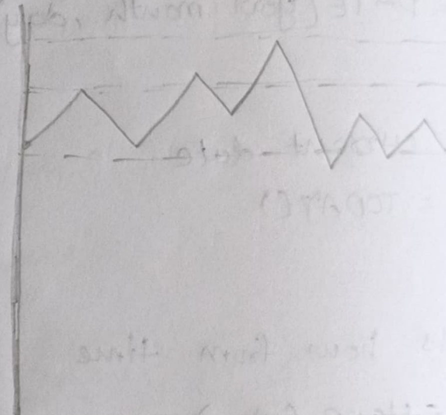
CHARTS

Why use charts?

→ Charts convert data into visual form, making it easier to
- Identify trends
- Compare values
- find patterns
- Detect outliers
- Present insights clearly

Use: Dashboards, reports, data analysis

Line chart



→ Used to show trends over time

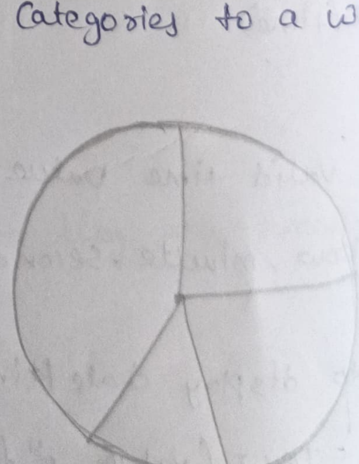
→ Best for:
- monthly sales
- Daily website visitors
- stock prices
- Job postings over time

→ advantage:
- Easy to identify trends
- shows growth & decline clearly

Pie chart

→ used to show percentage contribution of categories to a whole.

→ best when:
- few categories (3-6)
- showing proportions
- showing market share
- Showing category contribution

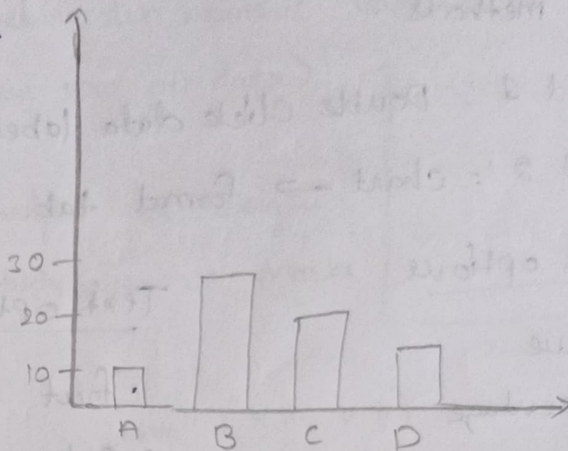


→ avoid when:
- Too many categories
- Compare similar values

Column Chart

→ uses vertical bars to represent values

→ best for : • Comparing categories
• showing difference b/w values



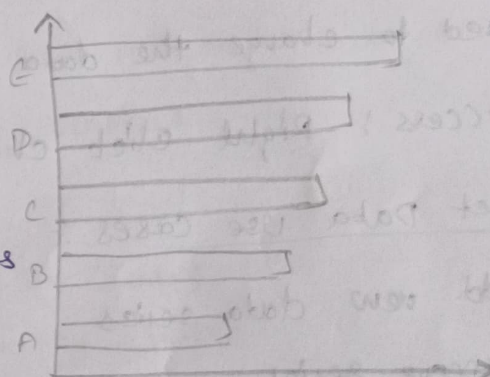
→ Ex:- sales of different products
Marks of different students

→ use when : • Category names are short
• No.of categories is small to medium
• You want an easy visual Comparision

Bar Chart

→ Uses horizontal bars to represent values

→ best for : • comparing many categories
• Ranking data
• showing top & bottom performers



→ Ex:- Top 10 students by marks
Top selling products

Chart Elements option [click chart then +]

→ used to improve chart readability

Common elements

- Chart Title
- Axis titles
- data labels
- Gridline
- Error bard
- Trendlines

Chart Design Tab

→ Used to modify chart appearance

→ Options include • Change chart Type • Quick layout • change colors • Move chart

Format Data labels

Access Methods :

Method 1 :- Double click data labels

Method 2 :- chart → format tab → format data labels

label options

- Value
- Percentage
- Category Name
- Series Name

Text options

- font
- color
- size
- Alignment

Select Data

→ Used to change the data displayed in the chart

→ Access : Right click chart & select 'select data'.

Select Data use cases :-

- add new data series
- remove series
- modify chart range
- change category labels

Select Data Source

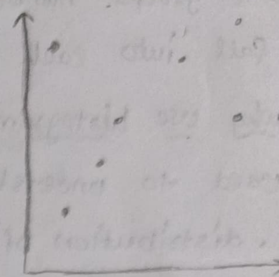
Controls
- chart Data Range
- series names
- Category labels

→ useful when chart is displaying incorrect data

Scatter Plots

→ used to show the relationship between two numeric variables

→ each dot represents one observation (one row of data)



why use scatter plots?

→ used to identify :
- relationships
- correlations
- patterns
- outliers

Ex:- height vs weight
Study hours vs Marks

→ scatter plots help identify correlation
- +ve (x↑ , y↑)
- -ve (x↑ , y↓)
- No (no visible patterns)

→ outliers

· scatter plots make unusual values easy to spot

Map chart

→ used to visualize data based on geographic locations

→ Excel automatically plots values on a map

why use map charts?

→ used to analyze data by :
- Country
- State
- City
- Region
- Postal code

Ex:- Sales by Country
Revenue by state

Note :- Chart Tips for Data Analysis

→ always add a chart Title

→ add data labels when important

→ sort data before creating charts

→ avoid unnecessary colors & effects

→ focus on insights , not decoration

Histogram

→ used to show the distribution of numeric data.

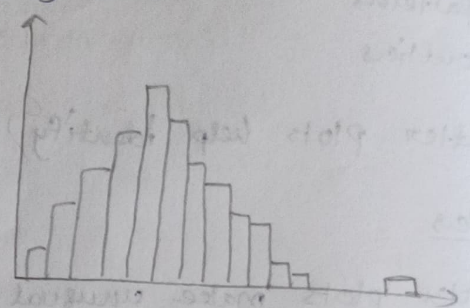
→ It groups numbers into ranges called bins & shows how many values fall into each range.

why use histogram?

Ex:- Exam scores

Employee salaries

Daily sales



used to understand

- distribution of data
- frequency of values
- Data spread
- skewness
- outliers

→ highest bar = most frequent values

→ spread = how widely data is distributed

→ skewness = whether data leans left or right

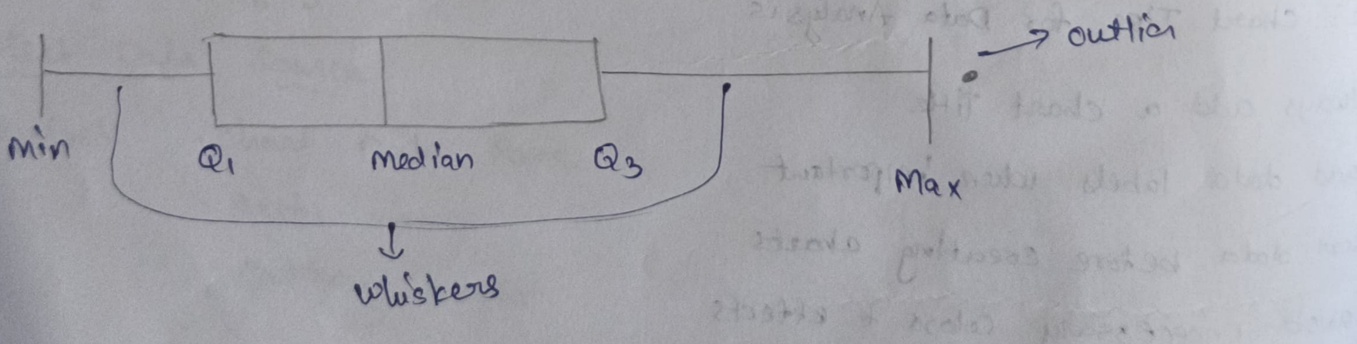
→ outliers = unusual values far from most data

Box & whisker chart

→ used to visualize the spread & distribution of data

shows :

- Minimum
- Q1 (1st Quartile)
- Median
- Q3 (3rd Quartile)
- Maximum
- Outliers



→ best for

- detecting outliers
- understanding spread
- comparing distributions

Combo chart

→ combines two different chart types in single chart

most commonly : column chart + line chart

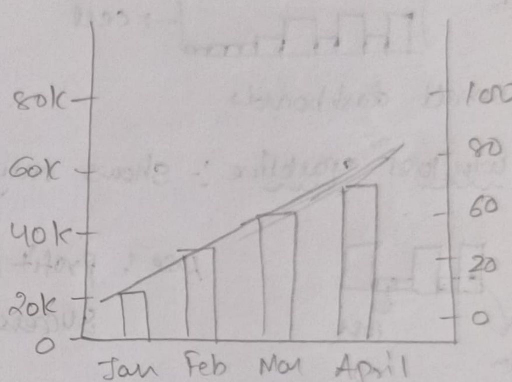
why use combo charts ?

→ used when comparing two related measures with different scales

ex:

Month	Revenue	Profit %
Jan	10000	10
Feb	20000	20
Mar	30000	30

Revenue is large, profit % is small



→ Combo chart makes both easy to see

Revenue → column chart

profit% → line chart

→ combo chart often uses a secondary axis

ex: left axis → Revence

Right axis → profit %

when to use • when comparing two metrics

• metrics have different scales

• showing trend + value together

Funnel chart

Shows how values decrease through stages of a process

Use: recruitment pipeline, sales funnel, website conversions

waterfall chart

shows how positive and negative changes contribute to a final value.

Use: Revenue → expenses → profit, Budget analysis

Sunburst chart

→ shows hierarchical data using concentric rings

Use: category → subcategory analysis

Treemap chart

→ shows hierarchical data using nested rectangles

Use: compare category sizes while also showing subcategories

Sparklines

→ Tiny chart that fits inside a single Excel cell.

→ Used to quickly show trends without taking much space

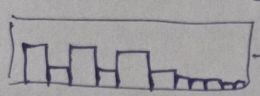
Types

1. line sparkline : shows trend over time

Ex: Jan → Feb → Mar → Apr

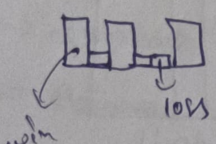
2. column sparkline : shows values as mini columns

Ex



→ small dashboards

3. win/loss sparkline :- shows only positive or negative results



Use : profit / loss

success / failure

Sparkline design tab

→ appears when a sparkline is selected

options : Sparkline color : change line color

Marker color : highlight points

High point : highlights highest point

low point : highlights lowest point

first point : highlights first point

last point : highlighs last point

Limitations

→ No axis labels

→ less detailed than normal charts

→ Not suitable for presentations.